



FACULTY OF COMPUTING
SEMESTER 1/20252026

SECR1013 – DIGITAL LOGIC

SECTION 03

PROJECT DIGITAL LOGIC

Electronic Lift/Elevator Controller System
(Simulation Using Deeds Software)

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1.0 Dedication & Acknowledgement

This final project is dedicated to our lecturer, Dr Raja Zahilah Binti Raja Mohd Radzi, who has guided us throughout the Digital Logic course. Her dedication, patience and clear explanation have greatly enhanced our understanding of digital logic concepts and their real-world application. This project stands as a reflection of the knowledge and skills we have gained under her guidance. In addition, we would also like to express our sincere appreciation to our group members for their commitment and cooperation throughout the project. Working on this final project has not only strengthened our understanding of Digital Logic, but also enhanced our critical thinking, problem-solving, and communication skills.

2.0 Objective

- To design and implement a secure password-protected lift system.
- To incorporate combinational circuit components such as comparators, decoders, multiplexers, and basic logic gates for input processing, floor comparison, and output control.
- To utilise sequential circuit elements including counters, flip-flops, latches, and logic gates to accurately control floor counting, movement direction, and door operations.
- To verify correct operation through simulation and testing in DEEDS.
- To showcase the application of digital electronics principles in the context of lift.

3.0 Background

In this project, we aim to develop a secure and interactive elevator control system using Digital Electronics Deeds Simulator. To activate the elevator, the user must first select the desired floor, then scan the access card, and finally enter the correct 4-bit passcode. Once authentication is successful, the clock will be enabled, and the counter will move the elevator to the selected floor. The design integrates combinational and sequential logic and adds additional security elements, such as an alarm and a locking mechanism after three incorrect password attempts.

3.1 Combinational circuit components

- i.) Basic Gate (AND, OR, NOT, NAND, XNOR)
- ii.) Input Switch
- iii.) Output Components
- iv.) Comparator
- v.) LEDs
- vi.) Seven Segment Display

3.2 Sequential circuit components

- i.) Clock Disabler
- ii.) Clock Enabler
- iii.) J-K Flip Flop
- iv.) S-R Latch

3.3 Extra Features

- i.) Door System (Open / Close)
- ii.) 4-bit Passcode System
- iii.) Access Card System
- iv.) Security Alarm System (Activates when enters wrong passcode 3 times)
- v.) Greeting System

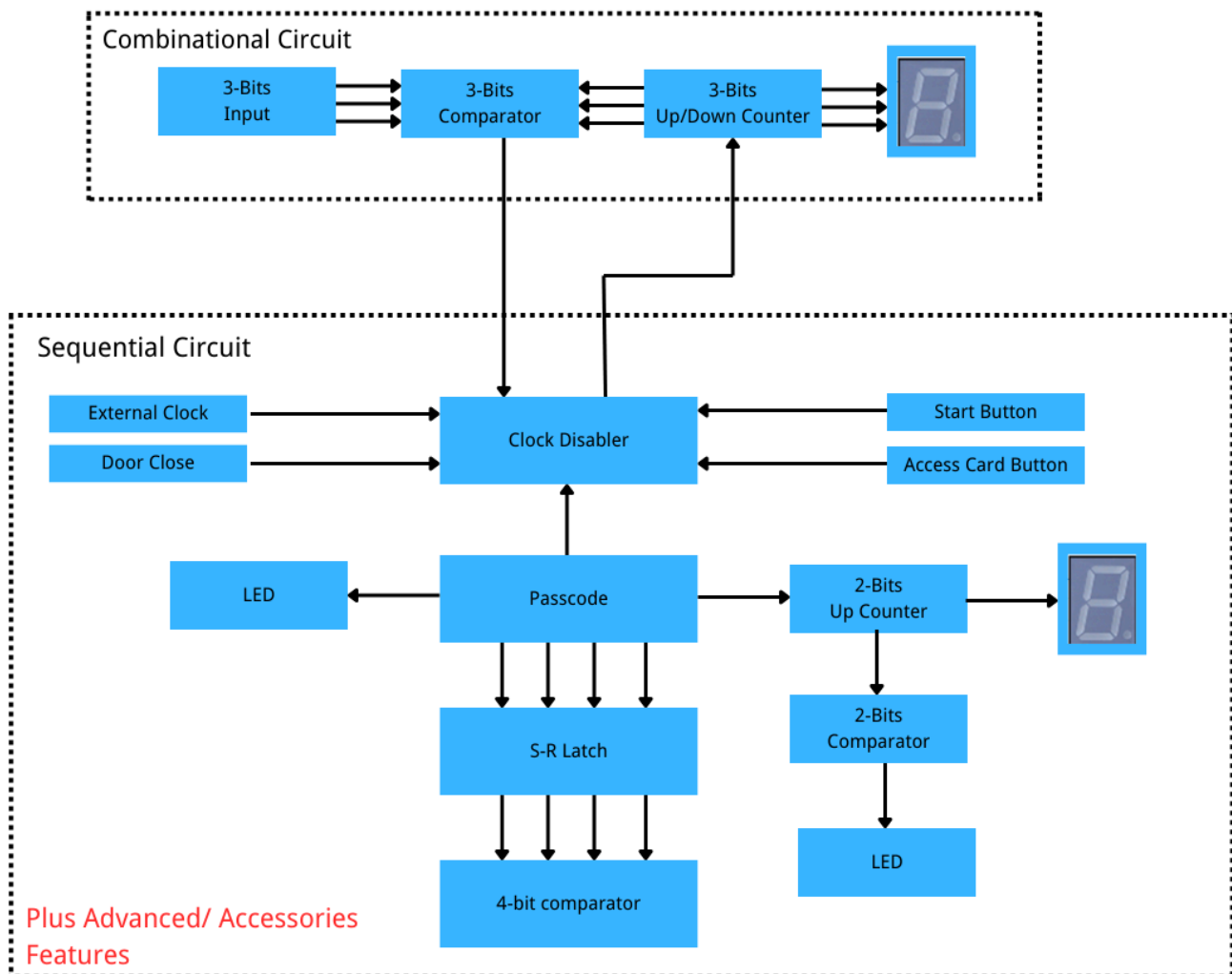
4.0 Problem Statement

Many elevators today operate without any security measures, which can lead to unauthorized use and safety risks. They also usually lack interactive features to guide users or alert them when something goes wrong. Because of this, it's important to have a lift system that is not only functional but also safe and user-friendly. By adding features like password protection, access cards, and alarms, we can make sure that only authorized people can use the lift and that the system reacts properly if something goes wrong. This project focuses on creating a secure and interactive lift system using the DEEDS simulator. It allows us to apply digital logic concepts practically, while also solving real-life problems of safety and accessibility in elevators.

5.0 Suggested Solution

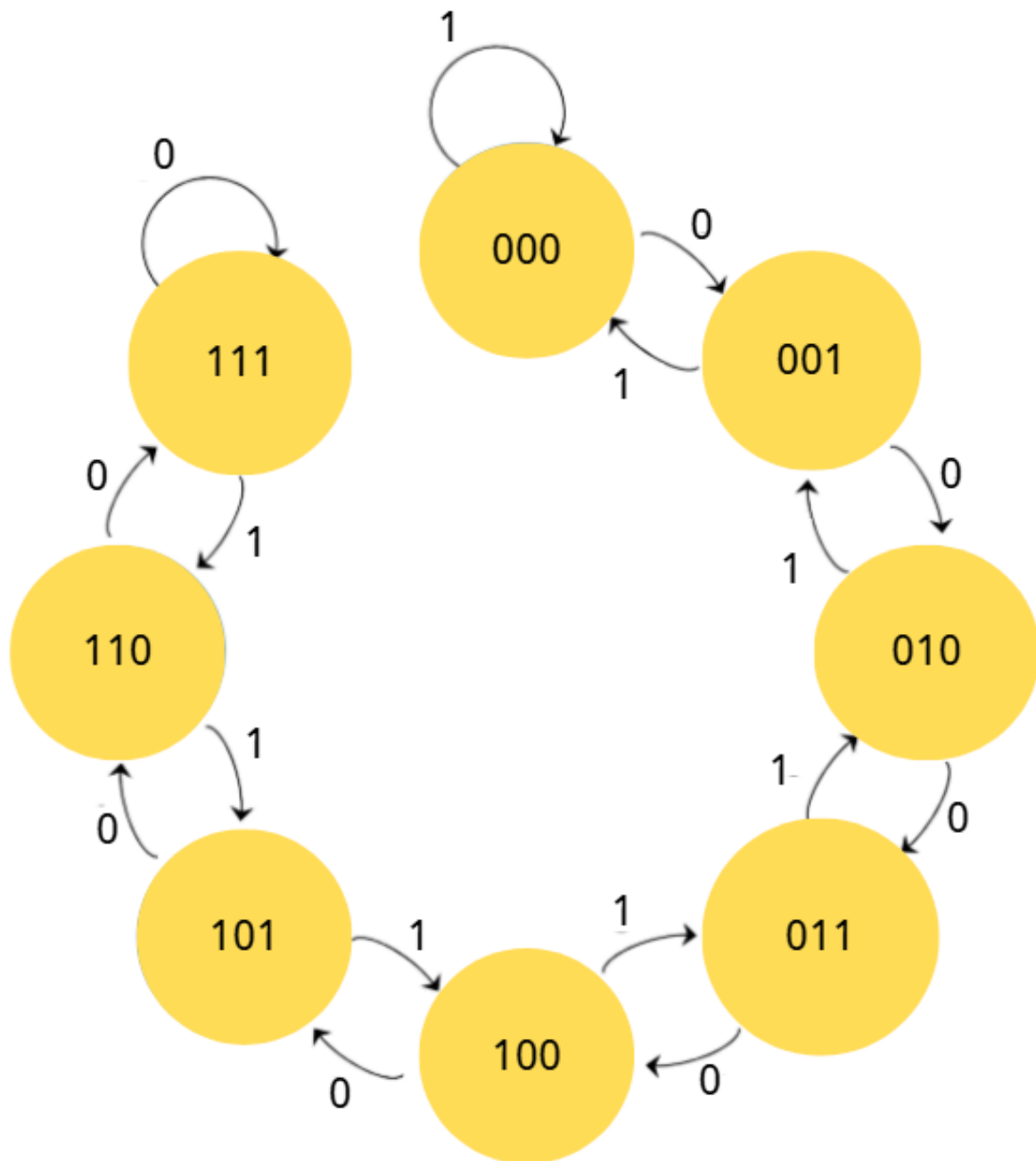
To address the issues of security and user interaction in elevator systems, we propose a secured and interactive lift design that integrates several digital components, including a 3-bit comparator, logic gates, switches, LEDs, JK flip-flops, an SR latch, and a clock control circuit, all implemented using the Digital Electronics DEEDS Simulator. In this system, the user must first select the desired floor, then scan the access card, and finally enter a correct 4-bit password before the lift is activated. Once authenticated, the greeting display ("HELLO") will appear, and the clock will be enabled, allowing the elevator to move according to the comparator output, which determines whether the lift should go up or down. A 3-bit counter tracks the current floor, and the lift stops automatically when the selected floor is reached. To strengthen security, an alarm and lockout mechanism is included, where three incorrect password attempts trigger the alarm and disable further operation. This proposed solution provides a reliable authentication system and ensures that only authorised users can operate the lift safely and efficiently.

5.1 Block Diagram



This block diagram shows the overall operation of the lift system. In the input blocks, we have the floor selection switches, the access for card input and the passcode input. For the comparator block, it will compare the current floor with the selected floor. Then it will determine whether the lift should move up, move down or stop. Clock control block enables the clock for lift movement and it will be disabled when destination is reached or system is locked. The last block is the output block which includes the movement of the lift, whether moving up or moving down, system for the close or open, alarm system and the display of the floor indication that used seven segment display LEDs.

5.2 State Diagram



Each state of the diagram shows one state of the lift. $X = 0$ shows the lift goes up and $X = 1$ shows the lift goes down. Transition happens when the selected floor is not the same as the current floor. It happens only when the up or down button is switched and the destination is achieved.

5.3 Next State and Transition Table

X	Present State			Next State			JK Flip Flop					
	Q2	Q1	Q0	Q2+	Q1+	Q0+	J2	K2	J1	K1	J0	K0
0	0	0	0	0	0	1	0	x	0	x	1	x
0	0	0	1	0	1	0	0	x	1	x	x	1
0	0	1	0	0	1	1	0	x	x	0	1	x
0	0	1	1	1	0	0	1	x	x	1	x	1
0	1	0	0	1	0	1	x	0	0	x	1	x
0	1	0	1	1	1	0	x	0	1	x	x	1
0	1	1	0	1	1	1	x	0	x	0	1	x
0	1	1	1	1	1	1	x	0	x	0	x	0

X	Present State			Next State			JK Flip Flop					
	Q2	Q1	Q0	Q2+	Q1+	Q0+	J2	K2	J1	K1	J0	K0
1	0	0	0	0	0	0	0	x	0	x	0	x
1	0	0	1	0	0	0	0	x	0	x	x	1
1	0	1	0	0	0	1	0	x	x	1	1	x
1	0	1	1	0	1	0	0	x	x	0	x	1
1	1	0	0	0	1	1	x	1	1	x	1	x
1	1	0	1	1	0	0	x	0	0	x	x	1
1	1	1	0	1	0	1	x	0	x	1	1	x
1	1	1	1	1	1	0	x	0	x	0	x	1

The table helps to determine the next state based on the current state, Q2, Q1, Q0 and the input X which is the up and down button. It is used to determine the input for JK flip-flop. The main function of this is to make sure the lift moves only one floor at one time and prevent the conflict of the state.

5.4 Karnaugh Map (K-Map)

Q1Q0 XQ2	Q1Q0			
	00	01	11	10
00	0	0	1	0
01	X	X	X	X
11	X	X	X	X
10	0	0	0	0

$$J_2 = \bar{X}Q_0Q_1$$

Q1Q0 XQ2	Q1Q0			
	00	01	11	10
00	X	X	X	X
01	0	0	0	0
11	1	0	0	0
10	X	X	X	X

$$K_2 = X\bar{Q}_1\bar{Q}_0$$

Q1Q0 XQ2	Q1Q0			
	00	01	11	10
00	0	1	X	X
01	0	1	X	X
11	1	0	X	X
10	0	0	X	X

$$J_1 = \bar{X}Q_0 + XQ_2\bar{Q}_0$$

Q1Q0 XQ2	Q1Q0			
	00	01	11	10
00	X	X	1	0
01	X	X	0	0
11	X	X	0	1
10	X	X	0	1

$$K_1 = \bar{X}\bar{Q}_2Q_0 + X\bar{Q}_0$$

Q1Q0 XQ2	Q1Q0			
	00	01	11	10
00	X	1	1	X
01	X	1	0	X
11	X	1	1	X
10	X	1	1	X

$$J_0 = \bar{X} + Q_2 + Q_1$$

Q1Q0 XQ2	Q1Q0			
	00	01	11	10
00	1	X	X	1
01	1	X	X	1
11	1	X	X	1
10	0	X	X	1

$$K_0 = X + \bar{Q}_1 + \bar{Q}_2$$

K-Map is used to simplify the Boolean expression to reduce the number of gates used in the circuit. It is very important to do this to make the implementation in the DEEDS much easier.

6.0 Implementation

This part explains how the main components of our lift system were implemented using the DEEDS simulator. Each component plays a specific role in making the lift secure, interactive, and functional.

6.1 Input Switch

Switches	Explanation
Floor selection	Users select the destination floor. The selected floor is represented in binary and sent to the comparator to be compared with current floor value
Access card	An access card acts as an enable signal for the system. The lift system will not operate unless this switch is turned ON to ensure that only authorised user can access the lift
Passcode input	Users need to enter a 4 bit password. The password is verified using logic gates. If the password is correct, the system enable the clock signal and lift begins operation otherwise the attempt counter increases and alarm triggered after three failed attempts

6.2 JK flip-flop

A JK flip-flop is used to store the state of the lift. This flip-flop controls the up and down movement of the lift at a time and the changing of the floor. This JK flip-flop is chosen because it is more flexible and compatible for the counter design. When the clock signal is active, the flip-flops will update their output. The counter increases when the lift moves up and decreases when the lift moves down at a time. The clock is enabled only after the access card and correct passcode entered. Once the lift reaches the selected floor, the clock is disabled and lift stops.

6.3 Comparator

The comparator in the floor sensor feedback module to verify whether the lift cabin has reached each intermediate floor during movement, preventing overshooting and ensuring precise stopping. It used to compare current floor value with selected floor to ensure the lift moves in the correct direction and stops at the selected floor. A 3 bit comparator is implemented using logic gates in DEEDS.

The comparator produces three main outputs:

- Current floor < Selected floor : Lift moves up
- Current floor > Selected floor : Lift moves down
- Current floor = Selected floor : Lift stops

Besides comparing the current floor with the selected floor, comparators are also used in other parts of the lift control system to ensure safe and accurate operation. For example, a comparator is used in the password verification module, where the 4-bit password entered by the user is compared with the stored predefined password. Only when both values match does the system allow access to the lift controls. Another comparator is used in the range checking circuit to ensure that the selected floor value is valid and not outside the allowed range. In the alarm system, a comparator is used to count and compare the number of incorrect password attempts with a preset limit, which in this project is 11 (3 times). When the number of wrong attempts reaches or exceeds this limit, the comparator activates the alarm output to alert the system of unauthorized access. These additional uses of comparators significantly improve the system's reliability, security, and overall performance.

6.4 Clock Disabler

We implement the Clock Disabler using a 6-input AND gate. This 6-input AND gate takes inputs from the start button, a signal from access card scanner a signal from the passcode verification, a signal from the floor comparator, an external clock signal and a signal from the door circuit to ensure it is closed. The Clock Disabler will only become operational when all five inputs are in the high state (1). The output of the 6-input AND gate is then linked to the clock input of the JK flip-flops. The 6-input AND gate is used to start and stop the operation of the lift when the user's selected floor number had arrived.

6.5 S-R Latch

An S-R (Set-Reset) latch is used in the lift access system to manage user authentication and control the lift operation based on the access card and passcode input. The S-R latch functions as a memory element, storing the current state of access until the system is reset, ensuring that unauthorized users cannot operate the lift.

Function in the system:

I. Set (S) Input:

The S input is triggered when a user scans their access card and enters the correct passcode. When both conditions are met, the S-R latch is set, producing a high output ($Q = 1$) that enables the lift operation.

II. Reset (R) Input:

The R input is used to reset the latch after the lift has completed its movement to the selected floor, or after a system timeout. Resetting the latch ensures that the lift system is ready for the next user.

III. Q Output:

The Q output of the latch serves as a control signal for other parts of the system, such as the Clock Disabler and JK flip-flops, allowing the lift to operate only when the user has been successfully authenticated.

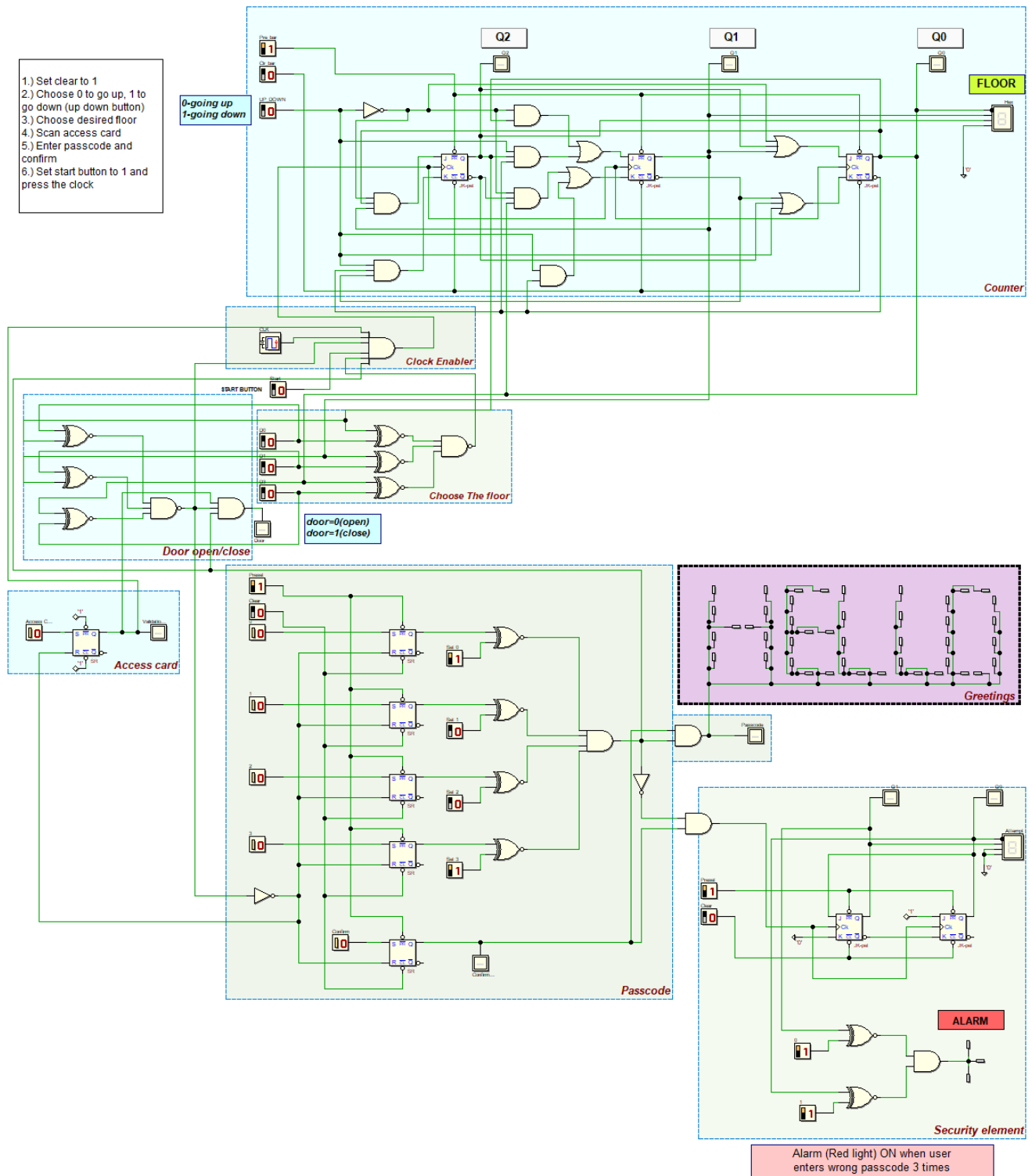
6.6 7-Segment Display

The 7-segment display is used in the lift control system to provide visual feedback to users and enhance system monitoring. It shows the current floor number of the lift cabin, allowing users to easily track its movement, and it also displays the number of incorrect passcode attempts in the alarm system. Each failed attempt increments the count on the display, and if the number of wrong attempts reaches the preset limit of three, the alarm is triggered to alert the system of unauthorized access.

6.7 LED Light

LED lights are used in the lift control system to provide visual signals for user interaction and security. A green LED displays a greeting, “HELLO,” when a user successfully authenticates with the access card and correct passcode, indicating that the lift is ready for operation. A red LED serves as an alarm indicator during lockout, lighting up when the number of incorrect passcode attempts reaches the preset limit, alerting users and the system to unauthorized access.

7.0 Full DEEDS Diagram



8.0 Conclusion and Reflection

Conclusion

In this project, we successfully designed and built a secure and interactive elevator system using the DEEDS simulator. The lift only operates after the user selects a floor, scans an access card, and enters the correct 4-digit passcode, which ensures both functionality and safety. We combined basic logic gates, comparators, switches, LEDs, and seven-segment displays with sequential elements, such as flip-flops, latches, and clock controls. This combination allowed the lift to move smoothly to the correct floor and handle inputs accurately. The extra features, such as the door open/close mechanism, greeting system, and security alarm, added realism and improved the overall functionality. In particular, the alarm and lock system that activates after three wrong password attempts highlighted the importance of safety in digital systems. Running and testing the system in DEEDS showed that our design worked as intended, reflecting our understanding of digital logic in a practical, real-world scenario.

Reflection

Working on this project was a valuable learning experience. We not only applied what we learned about combinational and sequential circuits, but we also saw how these components come together to form a complete system. Using flip-flops, latches, and clock controls helped us understand how timing and synchronization are essential for a working lift. One of the biggest challenges was making sure all the security features, such as the passcode, alarm and lock system worked correctly without interfering with the lift's movement. It took careful planning, testing, and teamwork to get everything right. Beyond technical skills, this project improved our teamwork, communication, and problem-solving abilities. Every group member contributed, and working together made the process smoother and more enjoyable. Overall, this project gave us more confidence in applying digital logic to real-world systems and prepared us for more complex projects in the future.

9.0 Reference

- [1]Foad, D. (2025, December 24). *Digital logic lab manual*. School of Computing, Faculty of Engineering, Universiti Teknologi Malaysia, Johor Bahru, Johor.
- [2]Abd. Bahrim Yusoff, Mazleena Salleh, M. F. Rohani, & Ismail Fauzi Isnin. (2025). *Digital logic* (5th ed.). Faculty of Computing, Universiti Teknologi Malaysia.
- [3]Widmer, N.S., Moss, G.L. and Tocci, R.J. (2017) *Digital Systems: Principles and applications*. Boston: Pearson.

10.0 Appendices

Name		HOWARD	TANESH	NOR AIN FAHIRA	NOR FATIHAH
Task	Designing	Full Circuit using Digital Electronics Simulator			
		Video Editing		Slide Preparation	
		State diagram, JK Flip Flop (Next state & Transition Table), K-map		Block Diagram	
	Report Writing	Acknowledgement and Dedication, Background, Objective	Implementation, Conclusion and Reflection	Suggested solution, Reference	Problem statement, Full DEEDS Diagram, Appendices

Group Members	Circuit
HOWARD LEE HAO ZHE	i.) Lift System(3 bit Counter, JK flip-flop) ii.) 4 bit Passcode Circuit iii.) Alarm Circuit
NOR AIN FAHIRA BINTI MUHAMAD FARIQ	i.) Door Circuit (Open / Close) ii.) Greeting Circuit iii.) Access Card Circuit
TANESH A/L K RAJAKUMARAN	i.) Lift System(3 bit Counter, JK flip-flop) ii.) 4 bit Passcode Circuit iii.) Alarm Circuit
NOR FATIHAH BINTI OTHMAN	i.) Door Circuit (Open / Close) ii.) Greeting Circuit iii.) Access Card Circuit

First Meeting

Date: 29/12/2025 (Monday)

Time: 9pm-10pm

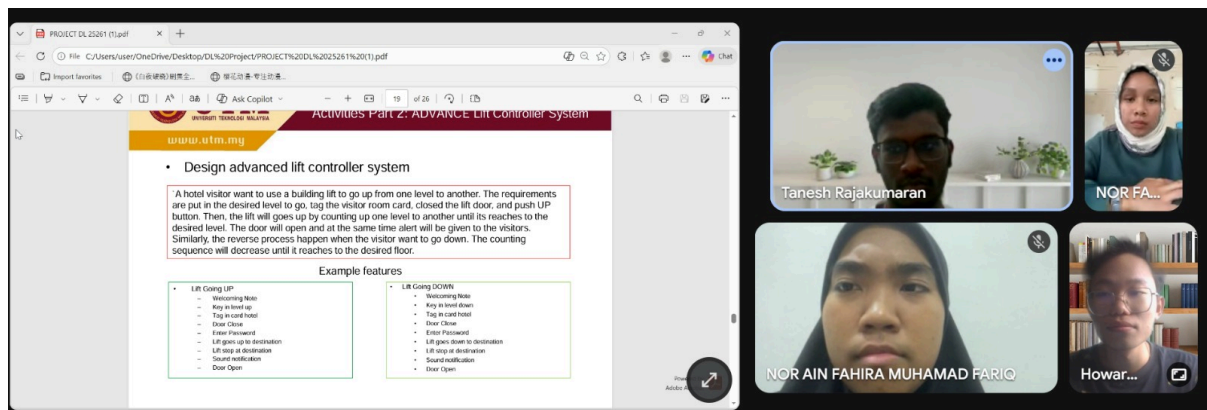
Venue: Online (Through Google Meet)

Attendance

1. HOWARD LEE HAO ZHE A25CS0067
2. NOR AIN FAHIRA BINTI MUHAMAD FARIQ A25CS0295
3. TANESH A/L K RAJAKUMARAN A25CS0359
4. NOR FATIHAH BINTI OTHMAN A25CS0296

Meeting Agenda:

1. Introduction of Group Members.
2. Project Discussion (Discuss on the additional features to add and distribute work to each member).
3. Scheduling of the Next Meeting
 - o Date: 3/1/2026 (Saturday)
 - o Time: 10pm



Second Meeting

Date: 3/1/2026 (Saturday)

Time: 10pm-11pm

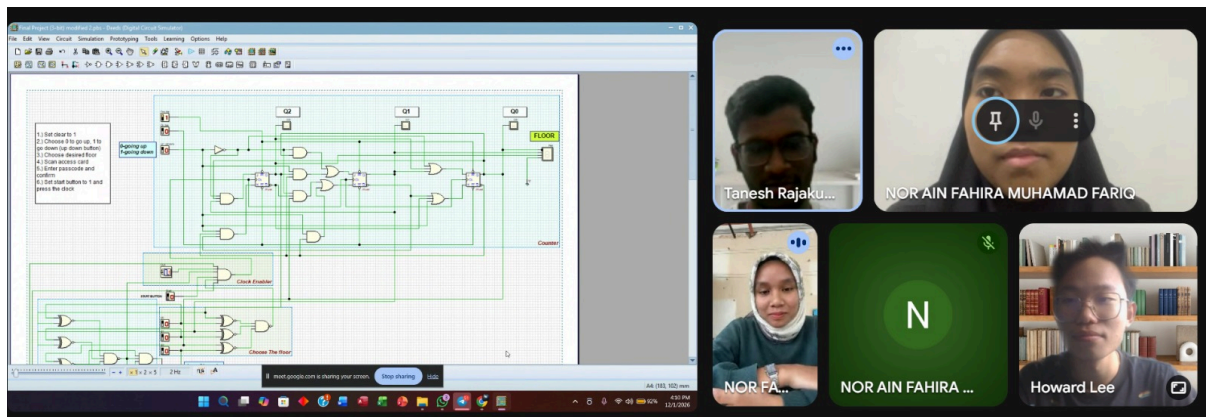
Venue: Online (Through Google Meet)

Attendance

1. HOWARD LEE HAO ZHE A25CS0067
2. NOR AIN FAHIRA BINTI MUHAMAD FARIQ A25CS0295
3. TANESH A/L K RAJAKUMARAN A25CS0359
4. NOR FATIHAH BINTI OTHMAN A25CS0296

Meeting Agenda:

1. Finalize the circuit with additional features and ensure it works without error.
2. Discussion on report distribution (report tasks are divided among members to ensure equal contribution and timely completion).
3. Scheduling of the Next Meeting
 - o Date: 10/1/2026 (Saturday)
 - o Time: 10pm



Third Meeting

Date: 10/1/2026 (Saturday)

Time: 10pm-11pm

Venue: Online (Through Google Meet)

Attendance

1. HOWARD LEE HAO ZHE A25CS0067
2. NOR AIN FAHIRA BINTI MUHAMAD FARIQ A25CS0295
3. TANESH A/L K RAJAKUMARAN A25CS0359
4. NOR FATIHAH BINTI OTHMAN A25CS0296

Meeting Agenda:

1. Finalize the report.
2. Discussion on slide preparation for video.
3. Planning for video.

The screenshot displays a Google Meet interface. On the left, a document titled 'report digital logic' is open, showing a table of contents and a section titled '8.2 State Diagram'. The diagram is a state machine with six states represented by yellow circles: 000, 001, 010, 011, 101, and 110. Transitions are labeled with 0 and 1. The right side of the screen shows four video thumbnails of participants: Tanesh Rajak... (top left), NOR AIN FAHIRA MUHAMAD FARIQ (top right), NOR FAT... (bottom left), and Howard Lee (bottom right).

Fourth Meeting

Date: 14/1/2026 (Wednesday)

Time: 9pm-10pm

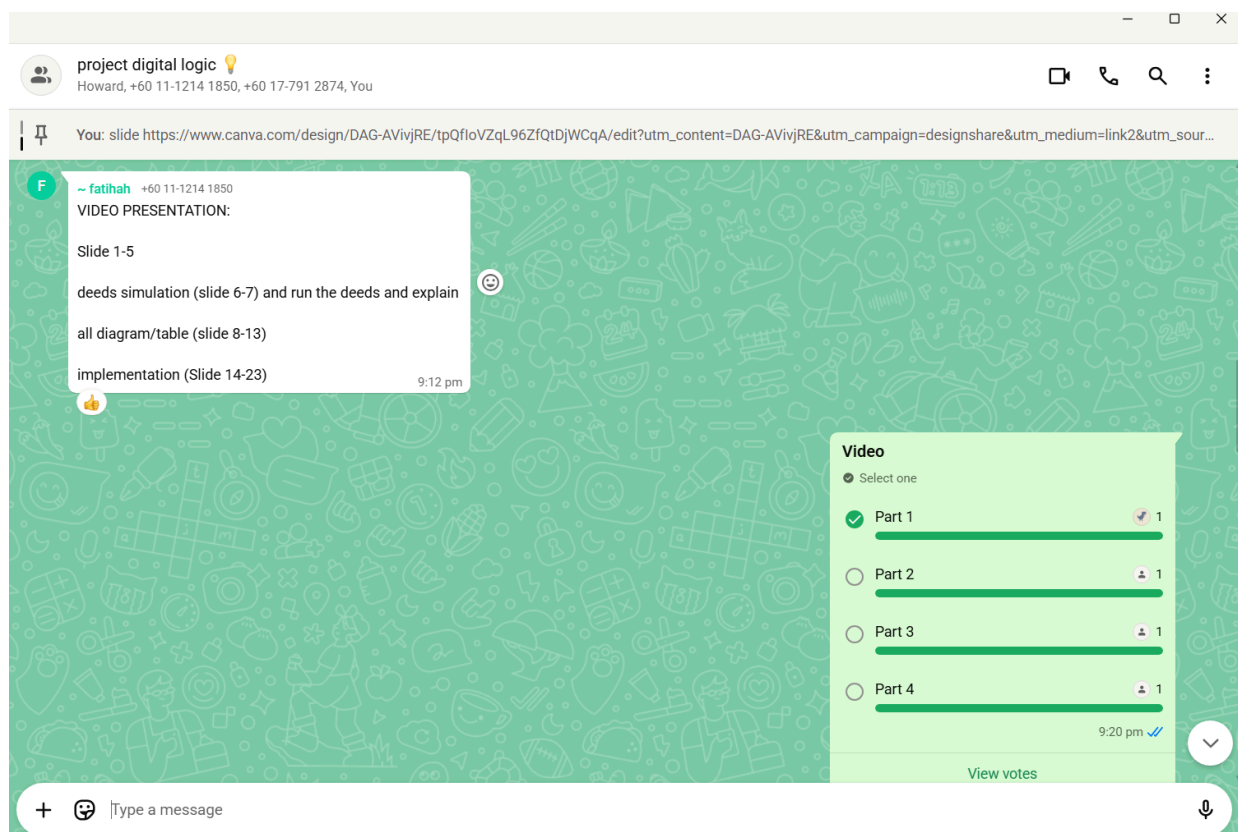
Venue: Online (Through Whatsapp)

Attendance

1. HOWARD LEE HAO ZHE A25CS0067
2. NOR AIN FAHIRA BINTI MUHAMAD FARIQ A25CS0295
3. TANESH A/L K RAJAKUMARAN A25CS0359
4. NOR FATIHAH BINTI OTHMAN A25CS0296

Meeting Agenda:

1. Finalize the slide.
2. Discuss on video (which part to present).



11.0 Video Link

<https://youtu.be/qi1j9YINMSs>